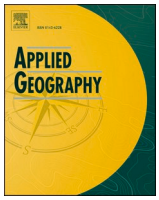
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Bringing economic complexity to the intra-urban scale: The role of services in the urban economy of Belo Horizonte, Brazil

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ABSTRACT

This study explores the formation of economic complexity within a city from the Global South, during 2011–2019. It proposes an expanded interpretation of the Economic Complexity Index (ECI) to be applied at the intra-urban context of Belo Horizonte, Brazil, focusing on three different spatial levels of analysis (i.e., local, neighbourhood, and community levels). By applying the index to these three levels, instead of regional or national administrative boundaries commonly used in literature, this study contributes to approximating the observation of economic complexity to the actual geographical scales at which economic interactions take place, allowing for intra-urban comparisons. The proposed ECI includes the service economy, amenities, and retail, in addition to commonly observed manufacturing industry. Methodologically, this case study introduces the Urban Economy Space network diagram to the expanded ECI as an effort to holistically consider all economic sectors happening in a city. The main findings are twofold. First, the city services classified as more complex by the ECI are aligned with the theory of post-industrial economic activities: financial, telecommunications, scientific and technical services, etc. Second, government-led institutions such as healthcare facilities, higher education institutions, etc., appear on the top tier of economic complexity, indicating that local and national governments can contribute to complexifying local economies.

1. Introduction

The process of innovation happens essentially within cities, when new labour is added on top of old labour and new business categories arise as breakaways from existing ones (Jacobs, 1970). This concept of innovation leads to the idea of economic development as a process being fairly linked to the diversification of a city's economic activities. Economic diversification also plays a role in enhancing a region's resilience, i.e., its capacity to withstand external shocks (Frenken, Van Oort, & Verburg, 2007; Xiao & Drucker, 2013; Sprague, 2018). Despite urban areas being crucial for economic development, current literature in geographies of innovation tends to focus on firms and entrepreneurial activities, at the regional or national scales (Florida, Adler, & Mellander, 2017), thus not usually addressing urban or intra-urban scales (Adler, Florida, King, & Mellander, 2019).

Recent economic geography literature points to economic complexity as a key ingredient for analysing an area's composition of economic activities (Bishop & Mateos-Garcia, 2019; Burlina & Antonietti, 2020). Economic complexity of countries is also positively associated with their levels of economic development (Hidalgo, 2015). As a measure at a national scale, the Economic Complexity Index (ECI) (Hidalgo & Hausmann, 2009) classifies products exported by countries, ranking their economies from more to less complex. The ECI has also been applied to regional economies (Bishop & Mateos-Garcia, 2019; Burlina & Antonietti, 2020), but is seldom used for understanding and comparing the economies of different areas within a city, nor applied to urban services, amenities, and retail, in extension to manufacturing industries or in place of exported products. This paper attempts to address this gap, applying the ECI to intra-urban areas, using as case study a city of the Global South.

1.1. Complexity and the city's economy

Economic diversification as a process is commonly seen as opposed

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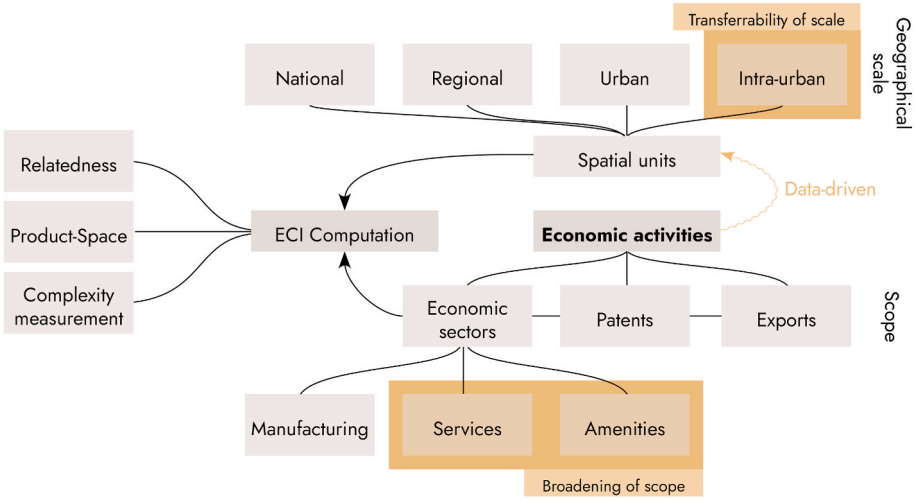


Fig. 1. Conceptual framework, highlighting the main contributions of this research to current literature in the topic.

to economic specialisation (Batty, 2017). While the former is considered a result of innovation led by knowledge spillovers in cities, the latter points out that agglomeration economies lead to ever-more specialised industries in a highly entrepreneurial environment (Burlina & Antonietti, 2020; Faggio, Silva, & Strange, 2020). The classical work of Jacobs (1970) credits the rise in economic diversity of cities to recurrent episodes of "import replacement". New industries may create new comparative advantages for products or processes in a city, where they did not previously happen, generating diversification. This is one of the reasons cities may abandon previous production processes, generating specialisation. This can lead to the emergence of related services and products, defined in literature as related variety (Bond-Smith & McCann, 2020; Frenken et al., 2007), that can again increase economic diversity and, in turn, foster further specialisation. These possible recurring processes of diversification and specialisation show that these two concepts do not necessarily oppose each other, but a successful economy may combine them in a harmonic way, leading to higher economic competitiveness in cities (Hong & Xiao, 2016).

Feedback processes across different scales, similar to the one described, is a key characteristic of complex systems studies (Salvati, Mavrakis, Serra, & Carlucci, 2015). Applied to whole economies, the study of economic complexity emerges as a relatively recent field, trying to link various socioeconomic phenomena to these non-linear dynamics of complex systems (Balland et al., 2022). Higher economic complexity has been linked to reduced greenhouse gas emissions (Mealy & Teytelboym, 2020; Romero & Gramkow, 2021), reduced shares of the shadow economy (Ha, Dung, & Thanh, 2021; Nguyen, 2022), reduced income and regional inequalities (Marco, Llano, & Pérez-Balsalobre, 2022; Zhu, Yu, & He, 2020), and a stronger income convergence between countries (Gala, Rocha, & Magacho, 2018). Bringing it closer in scale to the object of this research is possible since urban economies can also be characterised as complex systems, i.e., they are composed of heterogeneous agents in a variety of groups, acting in different times and spaces, incurring in non-linear patterns, generating unexpected outcomes (Burlina & Antonietti, 2020).

The idea of economic complexity in cities is also pervaded by the presence of knowledge-intensive, innovative products and services (Balland & Rigby, 2017). Large and well-connected cities tend to disproportionately concentrate innovation within its firms and production processes (Balland et al., 2020), usually associated with the post-industrial knowledge economy (Murdoch, 2018; Sassen, 2005). The presence of creative services in cities also contributes to higher employment growth and new business formation (Boschma & Fritsch, 2007). On the other hand, transition towards an intangible digital economy poses challenges for cities, such as rising structural unemployment with the gradual abandonment of mass-production systems (Bertani, Ponta, Raberto, Teglio, & Cincotti, 2021). Furthermore, Bettencourt, Lobo, Helbing, Kühnert, and West (2007) highlight the need for cities to continuously accelerate innovation cycles, essentially carried out within the sphere of knowledge-centred services and processes, in order to avoid stagnation or collapse. Addressing the digital transitions without leaving its citizens behind, while still avoiding the constant risk of stagnation and collapse, is a pressing dilemma for planners worldwide.

The knowledge-intensive service economy, essential for the rise and maintenance of this innovative environment, is embedded with consultancy firms, law firms, financial services, the creative sector, research and development (R&D). However, these as economic sectors are not often included in analyses of economic complexity. Other studies may include neighbourhood amenities (i.e., restaurants, hotels, bars, cafés) in analyses of economic relatedness and complexity (Hidalgo & Castañer, 2015; Hidalgo, Castañer, & Sevtsuk, 2020), but still leaving aside the previously mentioned service sectors, as well as retail. Therefore, there is a need for a holistic approach including multiple economic sectors happening within cities (manufacturing industries, the knowledge economy, amenities) to understand and study the entirety of a city's complex economic structure.

1.2. *Quantifying economic complexity: the ECI*

One way to quantify the complexity of a city's economic structure is by using the ECI (Hidalgo & Hausmann, 2009). The index originally assigns complexity values to both exported products and countries, representing the amount of crystallised knowledge involved in a product, and the presence of the right set of capabilities in a country for producing such products (Hidalgo, 2015). Since transmission of knowledge and capabilities has a strong geographic component, it has been described as being 'spatially sticky' (Balland & Rigby, 2017). Moreover, the ECI has also been applied to regional economies, by considering levels of employment in specific industry categories (Bishop & Mateos-Garcia, 2019; Burlina & Antonietti, 2020; Cicerone, McCann, & Venhorst, 2019), producing aggregated results for regional administrative boundaries. These aggregations, however, often fail to capture internal variations of the regions analysed, also ignoring key differences in local economic structures within the same city. By applying it to intra-urban areas these issues are addressed in this paper.

The seminal work regarding the Economic Complexity Index is Hidalgo and Hausmann (2009). From then, a series of multiple approaches have been implemented, having the field evolved in different directions, either towards studies of innovation via patent data, towards industrial



Fig. 2. Location of Belo Horizonte in Brazil.

organisation and industrial composition in a Schumpeterian approach (Sciarra, Chiarotti, Ridolfi, & Laio, 2020). Complexity measurements have been associated with economic performance (Gala et al., 2018), environmental performance (Romero & Gramkow, 2021), inequality (Marco et al., 2022). Refer to the work of Balland et al. (2022) for more detailed descriptions, references, and examples.

1.3. Research aims

The ECI is often applied to products exported by countries or regional labour structures, but seldom to urban or intra-urban economies. This research aims at understanding the structure of a city's economy, as measured by the ECI, and its derived relatedness matrices, and product-space networks, at different spatial scales. This allows to test the ECI's broadening of scope, by holistically considering all economic sectors in place of solely the commonly observed manufacturing industry; as well as the ECI's transferability of scale, by applying the index to intra-urban areas, using multi-scalar, data-driven spatial units.

1.4. Conceptual framework

The following conceptual framework (Fig. 1) summarises the current state-of-the-art of the literature in the topic, as well as highlights the main theoretical contributions of this research, beyond the shift of focus towards an emerging economy city. The presence of economic activities in a given (or chosen) geographical scale is the base object of analysis in any study about economic complexity, and the reading of this framework should start from it. However, the specific scope and object of analyses vary from field to field, or from research to research. It is common to find analyses that focus on exports (Poncet & Starosta de Waldemar, 2013), patent data (Balland, Boschma, Crespo, & Rigby, 2019), or the presence of certain economic sectors, such as occupational data from STEM categories (Lo Turco & Maggioni, 2022; Mealy & Teytelboym, 2020). The presence of economic activities by sector is often seen from the perspective of the manufacturing sectors (Dosi, Mathew, & Pugliese, 2022). The combination of these scopes with the spatial units chosen allows for the computation of the Economic Complexity Index, as well as all its derived measurements, such as relatedness matrices, the Product-Space, and the complexity measurement itself. At last, but not least, we consider the data-driven definition

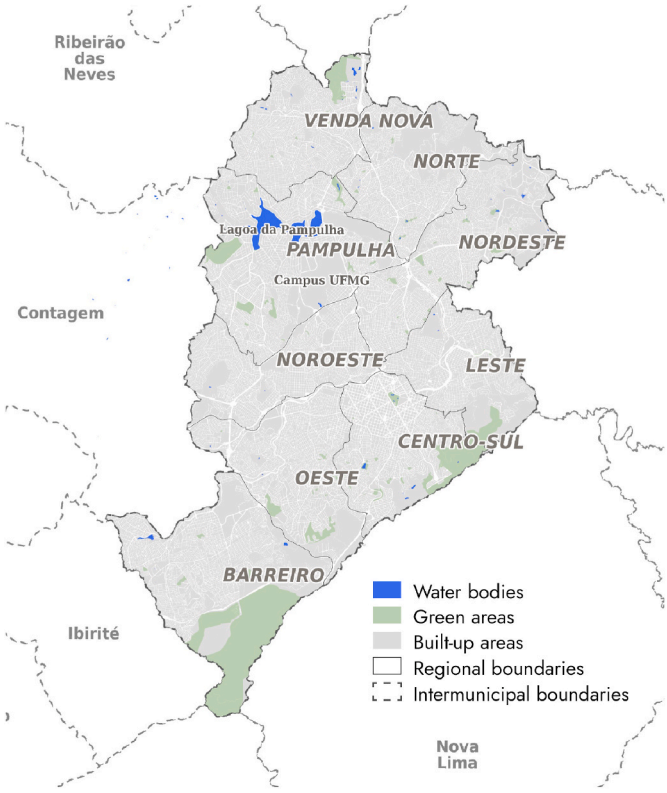


Fig. 3. Map of Belo Horizonte, highlighting intermunicipal and intra-municipal (regional) boundaries.

of spatial units to be an extra contribution of this work to the literature.

2. Methods

This section starts by defining the study area and the importance of such a study being done in its context. It follows with the description of the datasets used, and the delimitation of spatial units using a data-driven, bottom-up approach. Finally, it explains the calculation of the ECI index for the spatial units, as well as the methods used to analyse their internal composition of economic activities.

2.1. Study area

Belo Horizonte is the urban core of the third largest metropolitan area in Brazil, and is located in the heart of the country's iron ore mining region - the Iron Quadrangle (see Figs. 2 and 3). The city is the capital of Minas Gerais, a state historically dominated economically by the mining industry. In recent years, after two mining accidents totalling almost US $19 billion in economic losses and almost 300 fatal victims (Sapata Gonzalez, Aparecida da Silveira Rossi, & Gustavo Martins Vieira, 2022), not to mention an unpriceable environmental damage, growing discontent questioned the dominance of one single extraction industry in the state, with civil society pressure reaching transnational contexts (Cezne, 2019). In addition to being prone to deadly accidents, mining industry is expected to have a limited time-span, with mines reaching exhaustion and finally closing after a couple of decades. Dependency on mining also increases a place's vulnerability to external shocks, affecting labour market structures, households' income, and local governments' fiscal situations (Silva, Comini, Alves, da Rocha, & Jacovine, 2021).

Knowing that the mining industry is still growing, with mineral extraction expected to increase threefold to fourfold within the next decade (Coelho, Cordeiro, & Massola, 2020), understanding the current structure of Belo Horizonte's economy will be helpful to plan for its

future. Possible transitions towards a post-extractivist economy should include diversification strategies, human capital formation policies, and further public investment to increase local economic resilience (Silva et al., 2021). Reducing local dependency on mining can make use of complexification strategies at the local level, enabled by holistic environmental and social governance, already included in post-disaster mitigation efforts (Milanez, Ali, & Puppim de Oliveira, 2021) and in Brazilian legal framework in a broader sense (Haddad, 2015).

2.2. Data collection and filtering

The main dataset used for this research is a Vector – Point dataset with all economic activities registered in the municipality. Each point refers to a single economic activity, characterised by its official names, addresses, area used within a building, the type of activity, among others. The categorisation of the type of activities follows a nationally standardised coding system called CNAE (*Classificação Nacional de Atividades Econômicas* - National Classification of Economic Activities in Portuguese). This has been used in previous studies analysing the economic structure of Brazilian cities (Maraschin & Krafta, 2013; Barufi, 2018; Abreu, Del-Vecchio, & Grassi, 2020). It is the Brazilian equivalent of similar coding systems worldwide, such as the Netherlands' SBI (*Standaard Bedrijfsidentificatie* - Standard Business Identification in Dutch) (Smit, Abreu, & de Groot, 2015) and the U.S.'s North American Industrial Classification System (NAICS) (Sprague, 2018).

The dataset for the location of economic activities was available for the years of 2011, 2015 and 2019, allowing for temporal comparison throughout these 8 years. A thorough inspection of the mapped points did not detect mistakenly duplicate entries, mispositioned points or other inconsistencies. However, some degree of filtering was necessary for the purpose of the research. Three criteria were used: (1) whether economic activity's work is conducted within the registered and depicted address; (2) whether transaction or provider-customer contact is conducted at the depicted location; and (3) whether a location is characterised mainly by the presence of people (employers, employees, clients or others) in face-to-face contact (Storper & Venables, 2004) instead of machinery or goods. The filtering process included activities fulfilling at least one of these criteria and resulted in 199,407 entries for the year of 2019, 167,274 entries for the year of 2015 and 100,738 entries for the year of 2011.[1]

2.3. Spatial unit definition

The first step to define spatial units for the case study was to detect clusters of economic agglomeration by applying the Accessibility Index (Hidalgo & Castañer, 2015) for the filtered dataset of the most recent year (2019). The Accessibility Index (equation (1)) helped deriving the basic spatial units for this research. To be able to compare different years, we defined spatial units for the most recent year available and applied their boundaries to the datasets for the previous two years. This was done to avoid the usage of pre-determined administrative spatial units that aggregate characteristics to arbitrarily defined limits. This way, the concentration of economic activities by itself generates this study's spatial-analytical units.

Following Araldi and Fusco's (2019) multi-scalar approach for analysing urban services, three levels were defined: Local, Neighbourhood and Community. Equation (1) describes accessibility A of activity i, based on all other activities j, the distance d between i and j, and the constant γ used to detect three levels of clustering (Local, Neighbourhood and Community) by varying between 32, 16 and 8. For $\gamma = 32$, the influence of an activity (j) on another's (i) accessibility (A) decreases by half roughly every 21m, being neglectable at around 150m, defined here as the Local level of clustering. For $\gamma = 16$, the influence of an activity on another's accessibility decreases by half roughly every 42m, being neglectable at around 300m, defined here as the Neighbourhood level. For $\gamma = 8$, the influence of an activity on another's accessibility decreases by half roughly every 84m, being neglectable at around 600m, defined here as the Community level.

$$A_i = \sum_{j=1}^{N} e^{-\gamma d_{ij}} \tag{1}$$

For each level of aggregation, the accessibility values per activity are interpolated via an Inverse Distance Weighted (IDW) method, using an exponential function to weigh values by distance. The segmentation of the interpolated surfaces into units of analysis is conducted using SAGA's region-growing algorithm of Watershed Segmentation (Conrad et al., 2015). The algorithm selects local maxima as seeds, sectioning the area using the valley lines – i.e., areas with less concentration of economic activities – as edges, assigning for each spatial unit a unique peak of concentration to which it belongs. The result is a segmentation of the area into mutually exclusive spatial units polarised by points of high concentration of activities. To avoid oversegmentation, a peak-to-valley threshold equal to the raster's Standard Deviation was defined after a trial-and-error approach in line with the literature (e.g., Liu et al., 2018).

2.4. The economic complexity index

The ECI builds on two metrics: an area's Balassa Index (BI, Hidalgo, Klinger, Barabasi, & Hausmann, 2007; Hidalgo & Hausmann, 2009) and the Revealed Comparative Advantage (RCA). The BI is calculated by detecting whether a region has a higher share of a certain service than average (equation (2)). Here, the first differentiation of the usual application of the ECI: commonly, the BI is applied in trade flow analyses, detecting whether a certain product is more intensely exported by a region than average. The BI informs a binary RCA (equation (3)). By applying this to all economic sectors of the city, an area of a city having a higher number of a certain business category than the average of all areas ($BI > 1$) indicates that said area has an *RCA* in performing that category of service or retail. The intuition behind it is the following: if an area has a high concentration of, say, car dealerships, it is an indication that some specialisation has taken place, with the right set of capabilities in place for it to happen. From the point of view of trade theory, an analogy is also possible: higher concentrations of psychological services, for instance, is an indication that such area *exports* this service to other areas of the city, since an inflow of consumers is expected for that product or service from elsewhere in the city. Equation (2) contains a business category identifier (k), a spatial unit identifier (i), and the total number of areas (K). The number of firms (a) belonging to business category (k) present in spatial unit (i) is represented here as $a(k, i)$. A derived measurement used is the total number of firms for one specific business category in all areas, represented as a_i.

$$BI_{k,i} = \frac{a(k,i)}{a_i/K} \tag{2}$$

$$RCA_{k,i} = \begin{cases} 0, & \text{for } BI_{k,i} < 1 \\ 1, & \text{for } BI_{k,i} \geq 1 \end{cases} \tag{3}$$

By positioning the binary values for RCA in a proximity matrix with rows as spatial units and columns as business categories, the ECI calculation derives the diversity of an area as the number of business categories in which it has a comparative advantage, and the ubiquity of a category as the number of areas with a comparative advantage in it.

[1] It is important to highlight, at this point, that informal activities compose a high parcel of the Brazilian economy and are not included in this study due to the lack of data. Still, it is not believed to have influenced the results due to recent attempts by local and national governments to formalise individual entrepreneurs (e.g. domestic cooks, craftspersons, artists, etc.), who can become their own individual companies. Several entities of the sort were observed in the data.

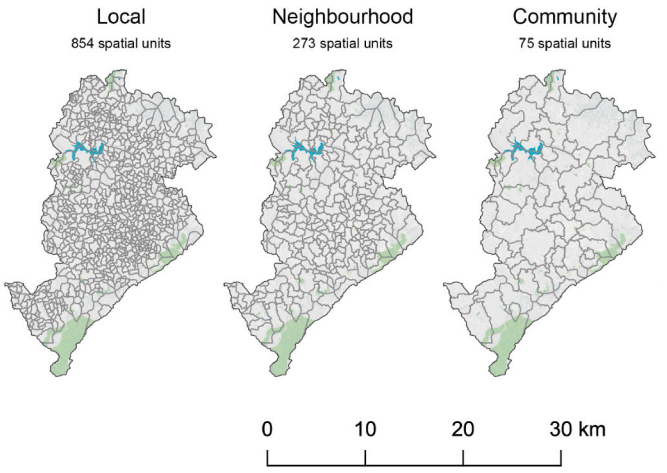


Fig. 4. Segmentation results defining the spatial units of this research for Belo Horizonte, Brazil.

Hidalgo and Hausmann (2009) define the calculation of the index via a so-called *method of reflections*: the previously calculated diversity of an area is updated according to its categories' ubiquities; the previous ubiquity of a category is updated according to their areas' updated diversities; the diversity is again updated according to the average diversity of the other areas with comparative advantage in the same categories; the ubiquity of a category is again updated according to the average ubiquity of the other categories within the same areas; and so forth, in an iterative process. A satisfying level of iteration (i.e., convergence) is also met by considering the eigenvector $\overrightarrow{C}$ with the second largest eigenvalue of the proximity matrix. In the equation below, $\langle \overrightarrow{C} \rangle$ represents the average and $std(\overrightarrow{C})$ represents the standard deviation. Economic Complexity Index (ECI – Equation (4)) for all areas k is then defined as follows:

$$ECI = \frac{\overrightarrow{C} - \langle \overrightarrow{C} \rangle}{std(\overrightarrow{C})} \tag{4}$$

BI, RCA and ECI were calculated using the R packages EconGeo (Balland, 2017) and EconomicComplexity (Vargas et al., 2020). Other derived products from these calculations are a complexity index for business categories assessing which categories were classified as complex, and proximity matrices for business categories, relating which categories tend to appear near one another. These are the base for generating the Product-Space network (Hidalgo et al., 2007), described in the following paragraph.

By using proximity matrices for business categories, it is possible to develop a network of categories based on the likelihood of them appearing within the same spatial unit, the Product Space (Hidalgo et al., 2007). The Product-Space gives meaningful insights on which business categories the ECI calculations are assigning as more or less complex. Its network form, with categories as nodes and probability of co-occurrence as edges, highlights how central nodes of economic activities are in relation to a whole network, indicating specific economic activities that enhance local complexity or fosters local diversification. This was detected by calculating the betweenness of this network's nodes.

A network of Product-Space was built for the lowest level of aggregation (Local level) with the highest number of spatial units, for the most recent year of analysis (2019). The resulting graph was analysed using igraph (Csardi & Nepusz, 2006) and characterised in relation to the business categories themselves (2-digit level - Fig. 7), the betweenness of nodes in the network, and the type of business categories assigned to the highest or lowest complexity values. Since this research's holistic usage of the Product-Space (Cicerone et al., 2019; Hidalgo et al., 2007) innovatively includes services in addition to manufacturing industries and amenities, considering the comparative advantage that business categories have by being more present than the average in each spatial unit, it was decided to give it the name of Urban Economy Space.

3. Results

The bottom-up, data-driven definition of spatial units generated 854 units for the Local level (i.e., the lowest level of clustering), 273 units for the Neighbourhood level, and 75 units for the Community level (Fig. 4).

Mapping the economic activities, detecting their concentration for different clustering levels, and segmenting the economic landscape into spatial units of analysis provide us already with interesting observations on how economic activities cluster in space. The Local level cluster detection for this research generated spatial units very close to the streetscape, being detected that the segmentations followed barriers also found in the landscape, such as larger street crossings, rivers, parks, among other areas with a lower concentration of economic activities. The same pattern is found to be reproduced for higher-level clustering. The Neighbourhood-level cluster detection is close in scale to the city's division of neighbourhoods, coinciding also with Regional boundaries when they exist. This can be seen in Fig. 5.



Fig. 5. Close-up view of Santa Tereza neighbourhood, divided into the three levels of clustering used in this research.

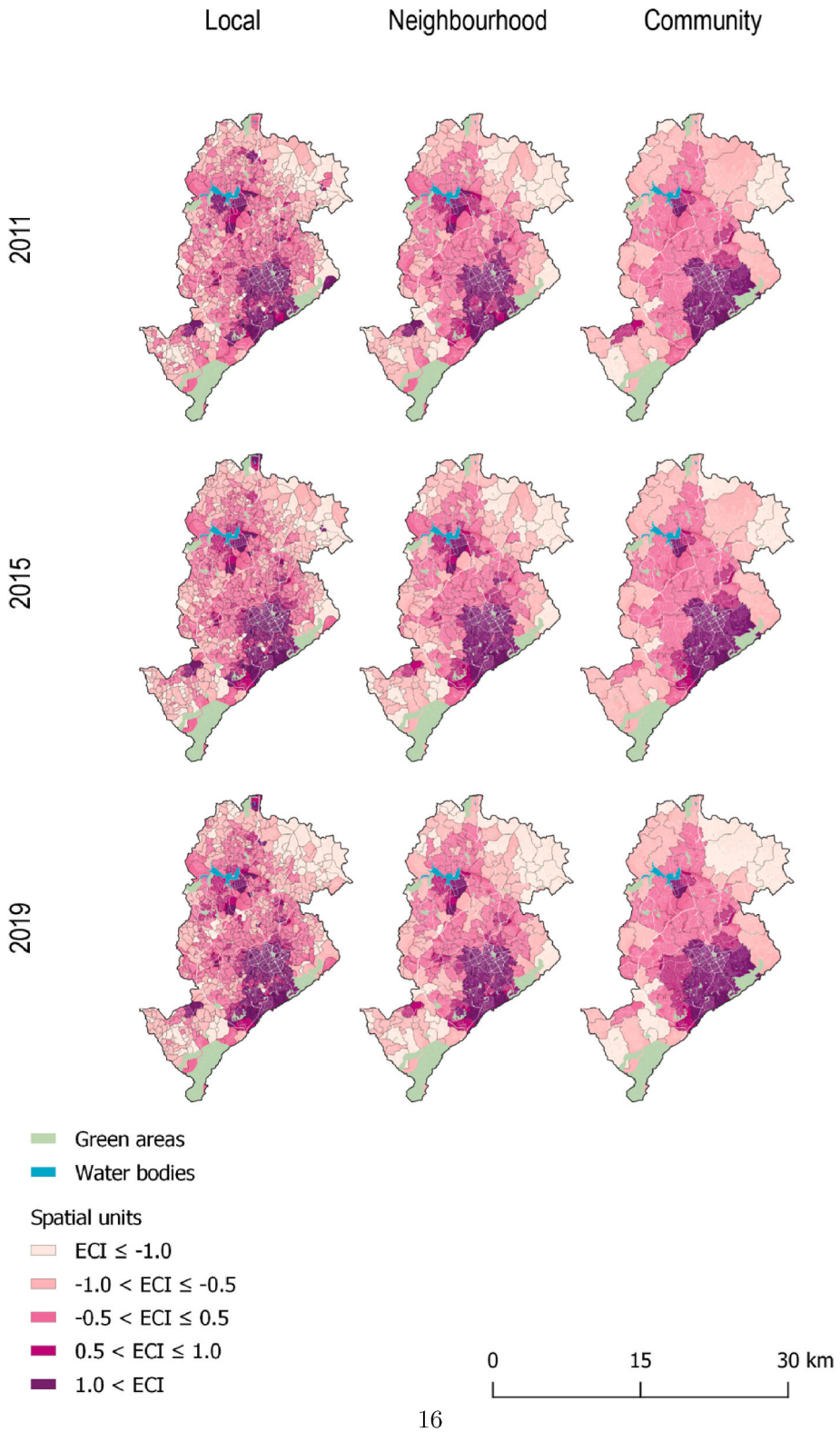


Fig. 6. ECI per spatial units in the different years and spatial scales.

3.1. *Economic structure of spatial units*

Fig. 6 shows ECI distributed in Belo Horizonte across scales and over time. In order to be able to compare different years, the chosen approach was to define the clusters for the most recent year available and apply their boundaries to the datasets for the previous two years. It can be seen that the highest ECI values are predominantly located in the original urban core of the city of Belo Horizonte across all scales. This is an area with orthogonal street layout, planned in the 1890s, and nowadays constitutes the wealthiest and most densified area of the city. Local level clustering also detected local commerce centres with higher ECI values, as well as the spatial unit comprising Minas Gerais state's Administrative

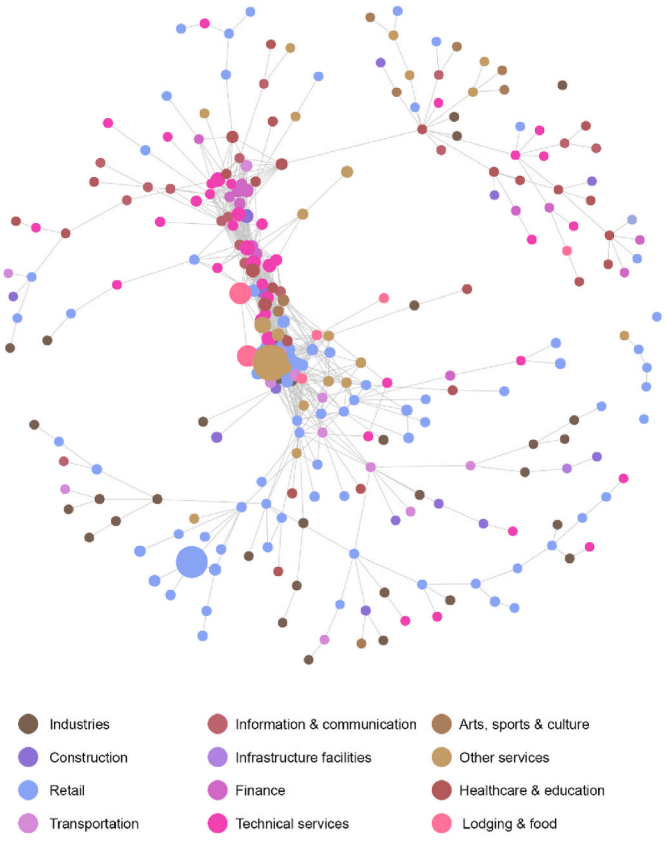


Fig. 7. The Urban Economy Space of the city of Belo Horizonte, categorised by 2-level CNAE aggregation of business categories. Circle sizes indicate number of firms in each business category.

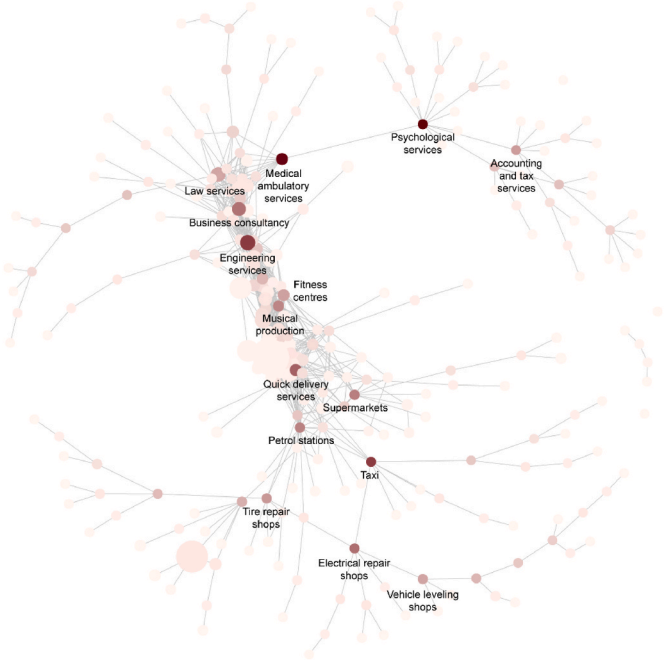


Fig. 8. Urban Economy Space classified by the betweenness of its nodes. Business categories with the highest betweenness are highlighted.

Headquarters, in the northernmost corner of Belo Horizonte – especially for years 2015 and 2019, after it was completed. The spatial units around the Federal University's campus also showed higher ECI values.

It is also important to understand how the ECI is related to the categorical composition of economic activities within these areas. The Urban Economy Space expands from the original Product-Space (Hidalgo et al., 2007) and the Amenity Space (Hidalgo & Castañer, 2015; Hidalgo et al., 2020) by considering all types of business categories, in place of exported products and in addition to neighbourhood amenities. By plotting these categories' co-occurrence in spatial units into a network, Fig. 7 shows a remarkable central core concentrating most of the economic categories observed. The branches located in the bottom corner of the chart, for instance, are more dedicated to retail activities, displaying the typical economic structure from neighbourhoods located in the global south.

The strength of the connection between these nodes is determined by the relevance of co-occurrence between two business categories in each spatial unit. This indicates that there are categories that tend to co-occur in space more often than others, generating nodes pervaded by many lines. The calculated betweenness for these nodes (Fig. 8) shows the business categories that tend to appear along most other business categories. These are not necessarily more complex, but their presence in a spatial unit indicate a more diversified economic landscape. These are detected to be pervasive categories of services either directed at households (e.g., fitness centres, vehicle repairing), or other businesses (e.g., accounting firms, consultancy).

The ECI calculations assign complexity values for both areas of analysis and business categories. To further investigate which areas are considered complex, the complexity of business categories is assessed. More (less) complex business categories are expected to require higher (lower) capabilities, higher (lower) technological requirements, or more (less) knowledge-intensive firms. Plotting the business categories with highest (Fig. 9) and lowest (Fig. 10) complexity values associated with them confirmed these expectations.

4. Discussion

4.1. The Urban Economy Space and ECI's transferability of scope

Firstly, the Urban Economy Space (Fig. 7) shows a network with a remarkable core of activities, indicating activities that tend to appear together in the spatial units. Contrary to the expectations (Hidalgo et al., 2007), this entangled network of firm category co-location does not necessarily translate into an enhanced economic complexity, or sophistication. The complexity of the business categories in this core ranges in mid-values, connecting branches of higher complexity in the top-right corner (Fig. 9) of the chart with branches of lower complexity in the bottom corner (Fig. 10). There are sub-networks of higher complexity and sub-networks of lower complexity, depending on the nature of economic activities that happen, and they both locate in the peripheries of the network.

Secondly, the analysis of betweenness in the network highlights that more generalised business categories act as catalysers for the presence of a higher number of categories and, therefore, a higher variety of activities. It can be concluded that the presence of categories such as law firms, consultancy firms, engineering services, supermarkets, fitness centres and music production, either fosters a richer composition of business categories within an area or is a consequence of it. In other words, the presence of these categories of firms is an indication of local economic diversity. This is an interesting finding since these categories do not appear to be themselves specialised. In fact, their level of generality – a law firm can act as auxiliary to a multitude of other business categories, for instance – may actually be the reason why they appear as strong catalysers in the first place.

The sort of business categories figured with the highest complexity values (Fig. 9) is in line with categories highlighted by Murdoch (2018) as demanding highly-skilled labour in the post-industrial economy and seen as the main sources of the economic prosperity for global cities (Sassen, 2005). These are financial, telecommunications, scientific and



Fig. 9. Urban Economy Space with highest ECI business categories.

technical services, educational services, healthcare and social assistance, among others. In this intra-urban analysis, these categories were also seen to contribute to overall higher levels of complexity of the economy. It is important to highlight that some of these categories are characterised by having a strong or exclusive governmental presence in the Brazilian context, especially medical services, higher educational facilities, hospitals, and public administration services (government-run by definition). This indicates that local and national governments can have a role in increasing the complexity of local economies.

On the other side of the spectrum, Murdoch (2018) describes less advanced industry categories as being economically traditional: construction, manufacturing, retail trade, transportation and warehousing, among others. These coincide with the lowest complexity categories observed in the Urban Economy Space (Fig. 10). This indicates a successful broadening in the scope when transferring applications of the Economic Complexity Index from the common country or regional level down to city and intra-city levels. Higher levels of aggregation (e.g. country-level), however, tend to position manufacturing industries within the higher complexity ones (Hidalgo, 2015), as opposed to commodity extraction and agriculture as low complexity. By transferring it to the city-scale, incorporating the service industry to the analysis, manufacturing figures among the lowest levels of complexity, possibly because agricultural and mining activities are even less commonly found in urban areas.

It is important to highlight, at this point, the nature of the calculation of the economic complexity index and how activities are assigned their own values of complexity. Since it is based on an imbalance by definition, as made explicit in the "method of reflections", when applying it to an intra-urban area some specific activities can be highlighted as complex, although not being considered in other situations. That was the case, specifically, for parking lots. Our guess is that some specific parts of the city, with a specific aspect of the urban form, are highlighted as complex, leading along the economic sectors that are part of them. Central, richer neighbourhoods were continuously highlighted as highly complex. Those central neighbourhoods, for reasons not necessarily isolated by this research, attract inhabitants from the whole Metropolitan area, and this might be reason enough for them to filled with parking lots, for instance. These richer neighbourhoods may, again for reasons not isolated by this research, attract more close attention from the public sector and be more often chosen for hosting a variety of public services, such as schools and clinics. This does not exclude the possibility of local governments complexifying local economies, but rather poses a question to the use of the method itself. However, as the index is created by a pre-defined imbalance, so are our cities, so is our society, and so are different countries. This may be a reason to dedicate attention to these issues, and the overall unequal nature of the economic system that shapes our environment and, by extension, our studies.

4.2. Research directions

The nature of the dataset was an important limitation for this research. Since locational data for firms is dependent on a formal registration with municipal authorities, informal economic activities, very relevant in the context of Brazil, are not depicted by the dataset. This is especially relevant for poorer areas, specifically the slum areas (favelas) and the poor peripheries typical for Brazilian cities. This means that the ECI values calculated for these areas are most likely lower than the actual complexity of the place, i.e. in case informal activities were considered. Further research in the field are encouraged to try to map and include informal activities.

It is also encouraged for further research to make a more explicit connection between common scale of applications of the ECI (i.e., regional and national scales) to the urban and intra-urban scales addressed in this paper. Similarly, the connections between processes or services detected at a local scale and exported products at a national scale can be made more explicit in further research. Moreover, this



Fig. 10. Urban Economy Space with lowest ECI business categories.

paper considers the mere location of activities in space as indication of local specialisation and, as such, the object of analysis. Given that capabilities and production processes are bounded geographically (Hidalgo, 2015), exploring local drivers of economic complexity, such as urban morphology indicators, is a natural development of this research.

Using higher granularity for spatial units, or even a spatial continuum of measurements in place of spatial units at all, could allow for a multi-level approach that prevents a significant Modifiable Areal Unit Problem (MAUP, Gotway & Young, 2002). In general, studies related to complex systems and emergent patterns should try to model individual elements and their relations to surrounding as close in scale as possible to the actual elements and their characteristics. That leads to a further recommendation for future studies that is to explore possible theoretical and microeconomic foundations for the complexity-enhancing process of recurrent diversification and specialisation, making use of simulations and agent-based models, for instance.

5. Conclusion

This research highlights that local or even national governments can have an important role in complexifying local economies. Some of the business categories classified as complex commonly belong to the public sphere, such as higher education facilities, hospitals, and public administration services. It is valuable that the positioning of these facilities in cities takes the underlying objective of fostering local economic complexity into account. Besides acting directly, governments could also (1) promote specific industry categories considered to be more complex, (2) enhance categories that are shown to be more connected to others (enhancing diversity), or, ideally, (3) prioritise multiple business categories that act together as networks of enhanced complexity. These are concluded by this research as potentially more efficient manners of promoting economic performance than the common directed specialisation policies, usually aiming at single sectors (Hong & Xiao, 2016). Overall, this research has made use of innovative methods and has contributed to enhance current literature in the field of economic complexity. The application of the Economic Complexity Index to intra-urban areas, including the service industry, bridges a gap between the economics of development and urban and regional sciences.

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Author contributions

Lucas Magalhães: Data curation, Formal analysis, Visualization, Writing - original draft, review & editing. **Monika Kuffer**: Project administration, Writing - original draft, review & editing. **Nina Schwarz:** Writing - original draft, review & editing. **Mônica Haddad:** Writing - original draft, review & editing.

Declaration of competing interest

The authors declare no conflicts of interest.

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References

Abreu, M. P., Del-Vecchio, R. R., & Grassi, R. (2020). Analysis of productive structure applying network theory: The Brazilian case. *Structural Change and Economic Dynamics, 53*, 281–291.

Adler, P., Florida, R., King, K., & Mellander, C. (2019). The city and high-tech startups: The spatial organization of Schumpeterian entrepreneurship. *Cities, 87*(December 2018), 121–130.

Araldi, A., & Fusco, G. (2019). Retail Fabric Assessment: Describing retail patterns within urban space. *Cities, 85*(September 2018), 51–62.

Balland, P.-A. (2017). *Economic geography in R: Introduction to the EconGeo package*. Utrecht: Technical report.

Balland, P.-A., Boschma, R., Crespo, J., & Rigby, D. L. (2019). Smart specialization policy in the European Union: Relatedness, knowledge complexity and regional diversification. *Regional Studies, 53*(9), 1252–1268.

Balland, P.-A., Broekel, T., Diodato, D., Giuliani, E., Hausmann, R., O'Clery, N., et al. (2022). The new paradigm of economic complexity. *Research Policy, 51*(3), Article 104450.

Balland, P.-A., Jara-Figueroa, C., Petralia, S. G., Steijn, M. P. A., Rigby, D. L., & Hidalgo, C. A. (2020). Complex economic activities concentrate in large cities. *Nature Human Behaviour, 4*(3), 248–254.

Balland, P.-A., & Rigby, D. (2017). The geography of complex knowledge. *Economic Geography, 93*(1), 1–23.

Barufi, A. M. B. (2018). Services that add value in the city: The rise of the modern economy in Brazil. *Cities, 78*(February), 39–51.

Batty, M. (2017). Diverse cities, successful cities. *Nature Human Behaviour, 1*(22), 2.

Bertani, F., Ponta, L., Raberto, M., Teglio, A., & Cincotti, S. (2021). The complexity of the intangible digital economy: An agent-based model. *Journal of Business Research, 129*, 527–540.

Bettencourt, L. M. A., Lobo, J., Helbing, D., Kühnert, C., & West, G. B. (2007). Growth, innovation, scaling, and the pace of life in cities. *Proceedings of the National Academy of Sciences, 104*(17), 7301–7306.

Bishop, A., & Mateos-Garcia, J. (2019). Exploring the link between economic complexity and emergent economic activities. *National Institute Economic Review, 249*(1), R47–R58.

Bond-Smith, S. C., & McCann, P. (2020). A multi-sector model of relatedness, growth and industry clustering. *Journal of Economic Geography, 20*(5), 1145–1163.

Boschma, R. A., & Fritsch, M. (2007). *Creative class and regional growth - empirical evidence from eight European countries*. Rochester, NY: Social Science Research Network. SSRN Scholarly Paper ID 1023869.

Burlina, C., & Antonietti, R. (2020). *From variety to economic complexity: Empirical evidence from Italian regions*. Utrecht: Technical report.

Cezne, E. (2019). Forging transnational ties from below: Challenging the Brazilian mining giant Vale S.A. Across the South Atlantic. *The Extractive Industries and Society, 6*(4), 1174–1183.

Cicerone, G., McCann, P., & Venhorst, V. A. (2019). Promoting regional growth and innovation: Relatedness, revealed comparative advantage and the product space. *Journal of Economic Geography, 20*(1), 293–316. https://doi.org/10.1093/jeg/lbz001

Coelho, P. S., Cordeiro, M. P., & Massola, G. M. (2020). Effects of female political participation on insecurity in a mining area. *International Perspectives in Psychology: Research, Practice, Consultation, 9*(1), 18–35.

Conrad, O., Bechtel, B., Bock, M., Dietrich, H., Fischer, E., Gerlitz, L., et al. (2015). System for automated geoscientific analyses (SAGA) v. 2.1.4. *Geoscientific Model Development, 8*(7), 1991–2007.

Csardi, G., & Nepusz, T. (2006). The igraph software package for complex network research. *InterJournal **Complex Sy***, 1695.

Dosi, G., Mathew, N., & Pugliese, E. (2022). What a firm produces matters: Processes of diversification, coherence and performances of Indian manufacturing firms. *Research Policy, 51*(8), Article 104152.

Faggio, G., Silva, O., & Strange, W. C. (2020). Tales of the city: What do agglomeration cases tell us about agglomeration in general? *Journal of Economic Geography, 20*(5), 1117–1143.

Florida, R., Adler, P., & Mellander, C. (2017). The city as innovation machine. *Regional Studies, 51*(1), 86–96.

Frenken, K., Van Oort, F., & Verburg, T. (2007). Related variety, unrelated variety and regional economic growth. *Regional Studies, 41*(5), 685–697.

Gala, P., Rocha, I., & Magacho, G. (2018). The structuralist revenge: Economic complexity as an important dimension to evaluate growth and development. *Brazilian Journal of Political Economy, 38*, 219–236.

Gotway, C. A., & Young, L. J. (2002). Combining incompatible spatial data. *Journal of the American Statistical Association, 97*(458), 632–648.

Haddad, M. A. (2015). A framework for urban environmental planning in Brazil. *European Review of Latin American and Caribbean Studies/Revista Europea De Estudios Latinoamericanos y Del Caribe*, (99), 113–125.

Ha, L. T., Dung, H. P., & Thanh, T. T. (2021). Economic complexity and shadow economy: A multi-dimensional analysis. *Economic Analysis and Policy, 72*, 408–422.

Hidalgo, C. A. (2015). *Why information grows: The evolution of order, from atoms to economies*. Penguin Books.

Hidalgo, C. A., & Castañer, E. E. (2015). *The amenity space and the evolution of neighborhoods*. arXiv. preprint arXiv:1509.02868 (2015).

Hidalgo, C. A., Castañer, E., & Sevtsuk, A. (2020). *The amenity mix of urban neighborhoods*. *Habitat International* (July).

Hidalgo, C. A., & Hausmann, R. (2009). The building blocks of economic complexity. *Proceedings of the National Academy of Sciences, 106*(26), 10570–10575.

Hidalgo, C. A., Klinger, B., Barabasi, A.-L., & Hausmann, R. (2007). The product space conditions the development of nations. *Science, 317*(5837), 482–487.

Hong, S., & Xiao, Y. (2016). The influence of multiple specializations on economic performance in U.S. Metropolitan areas. *Sustainability, 8*(9), 1–16.

Jacobs, J. (1970). *The economy of cities* (1st ed.). New York: Vintage Books.

Liu, L., Peng, Z., Wu, H., Jiao, H., Yu, Y., & Zhao, J. (2018). Fast identification of urban sprawl based on K-means clustering with population density and local spatial entropy. *Sustainability, 10*(8).

Lo Turco, A., & Maggioni, D. (2022). The knowledge and skill content of production complexity. *Research Policy, 51*(8), Article 104059.

Maraschin, C., & Krafta, R. (2013). Growth dynamic of retail locations: A methodological approach using a logistic model. *Cybergeo 2013*.

Marco, R., Llano, C., & Pérez-Balsalobre, S. (2022). Economic complexity, environmental quality and income equality: A new trilemma for regions? *Applied Geography, 139*, Article 102646.

Mealy, P., & Teytelboym, A. (2020). *Economic complexity and the green economy*. Research Policy.

Milanez, B., Ali, S. H., & Puppim de Oliveira, J. A. (2021). Mapping industrial disaster recovery: Lessons from mining dam failures in Brazil. *The Extractive Industries and Society, 8*(2), Article 100900.

Murdoch, J. (2018). Specialized vs. diversified: The role of neighborhood economies in shrinking cities. *Cities, 75*, 30–37.

Nguyen, C. P. (2022). Does economic complexity matter for the shadow economy? *Economic Analysis And Policy, 73*, 210–227.

Poncet, S., & Starosta de Waldemar, F. (2013). Export Upgrading and growth: The prerequisite of domestic Embeddedness. *World Development, 51*, 104–118.

Romero, J. P., & Gramkow, C. (2021). *Economic complexity and greenhouse gas emissions* (Vol. 139). World Development, Article 105317.

Salvati, L., Mavrakis, A., Serra, P., & Carlucci, M. (2015). Lost in translation, found in entropy: An exploratory data analysis of latent growth factors in a Mediterranean city (1960–2010). *Applied Geography, 60*, 107–119.

Sapata Gonzalez, R., Aparecida da Silveira Rossi, R., & Gustavo Martins Vieira, L. (2022). Economic and financial consequences of process accidents in Brazil: Multiple case studies. *Engineering Failure Analysis, 132*, Article 105934.

Sassen, S. (2005). The global city: Introducing a concept. *Brown Journal of World Affairs, 11*(2), 27–43.

Sciarra, C., Chiarotti, G., Ridolfi, L., & Laio, F. (2020). Reconciling contrasting views on economic complexity. *Nature Communications, 11*(1), 3352.

Silva, L. B., Comini, I. B., Alves, E. B. B. M., da Rocha, S. J. S. S., & Jacovine, L. A. G. (2021). Compensating the negative environmental impacts of mining with financial mechanisms in Brazil. *Land Use Policy, 104*, Article 105351.

Smit, M. J., Abreu, M. A., & de Groot, H. L. (2015). Micro-evidence on the determinants of innovation in The Netherlands: The relative importance of absorptive capacity and agglomeration externalities. *Papers in Regional Science, 94*(2), 249–272.

Sprague, C. D. (2018). *Resilience and the U.S. Labor market: A cross-scale analysis on the role of industrial diversity and specialization, PhD thesis*. Boise State University.

Storper, M., & Venables, A. J. (2004). Buzz: Face-to-face contact and the urban economy. *Journal of Economic Geography, 4*(4), 351–370.

Vargas, M., Bottai, C., Kozlowski, D., Pintar, N., Bank, T. W., & Statistics, O. T. (2020). *Computational methods for economic complexity*.

Xiao, Y., & Drucker, J. (2013). Does economic diversity enhance regional disaster resilience? *Journal of the American Planning Association, 79*(2), 148–160.

Zhu, S., Yu, C., & He, C. (2020). Export structures, income inequality and urban-rural divide in China. *Applied Geography, 115*, Article 102150.